Indian Institute of Information Technology, Surat

B.Tech. II (ECE & CSE), Semester – III

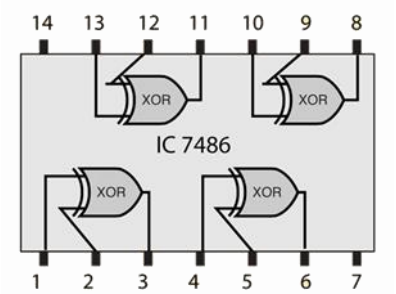
Microprocessor and Interfacing (EC 302)

End Semester Exam

Date: 14/12/2021 **Time: 10:40AM – 12:30PM** *Marks: 30*

Instructions

- *Attempt all questions.*
- *Follow all the guidelines given by institute exam-section while attempting the exam.*
- *The scanned pdf must have the file name as '**enrollment-no.pdf**'. For e.g. **ui20cs01.pdf**, **ui20ec01.pdf**……and so on.*

No.		Marks
1	a) Interface two 8K × 8 EPROMs (2764) with the 8086 using logic gates, such that the memory address ranges assigned to them are 20000H – 23FFFH.	3
	b) Interface and 8-bit output port having an address of 0FH with the 8086 microprocessor. Assume 8 LEDs are connected at the output port.	2
2	Draw the functional block diagram of 8086 microprocessor and explain its flag register in detail. Write example instructions to set the status flags. Show the necessary calculations/data-operations.	5
3	a) Write a time delay program to generate a delay of 10 minutes in an 8086 based system that runs on a 1 MHz frequency clock. Use the following instructions: MOV (4 T-state), NOP (3 T-state), DEC (2 T-state), JNZ (16/4 T-state), RET (8 T-state)	3
	b) Explain MUL and DIV instructions with examples.	2
4	Write an 8086 assembly language program to accept user-string as input. Find its length and display it on monitor screen. Length computation should be written as a procedure which will be called from the main-line program. A neat & clean flowchart demonstrating your logic is mandatory.	5
5	(a) Design an 8085 interfacing circuit to interface 8255 with 7486 IC (i.e. Quad 2-input Ex-OR gate) using 74138, appropriate gate/s so that PA address is D7H and other port addresses are based on this address. (b) for this circuit, write an 8085 Assembly Language Program (ALP) to check faulty gate (if any). 	5
6	a) Write breakpoint subroutine using 8085 RST 4 software interrupt such that whenever RST 4 instruction is placed it breaks the current program execution and move to Interrupt Service Routine (ISR) of RST 4. Implement RST 4 ISR such that it turn ON (logic 1) LEDs connected D7, D5, D3 and D1 of data bus for 1 second and turn it OFF 1 sec then return to main program. Assume 1 sec delay routine DELAY is available.	3
	b) Write the command word and indicate the contents of the command word that is required to configure the 8259 for the following operations: • ICW4 is going to be used • Three 8259s are to be used in the system such that the Slave 8259s are connected on IR_2 and IR_5 inputs of Master 8259 • The address interval is required to be 4 bytes • The IR_{0-7} requests are to be level triggered • The least significant byte address for IR_0 to be C0H	2

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